

Detecting Deforestation from Multi-Temporal Sentinel-2 Imagery: A Comparison of Transfer-Learning Land-Cover Classification and NDVI Differencing Validated Against Global Forest Watch

D A S K R Manogna

Dept. of Computer Science and Engineering
Indian Institute of Technology Gandhinagar
Gujarat, India

Divisht Dahiya

Dept. of Computer Science and Engineering
Indian Institute of Technology Gandhinagar
Gujarat, India

Abstract—Deforestation monitoring increasingly relies on automated analysis of satellite imagery. A natural approach is to train a land-cover classifier and apply it across two dates, flagging tiles whose class changes from forest to a non-forest category. We test this hypothesis rigorously. A ResNet-50 classifier fine-tuned on the EuroSAT dataset attains 98.61% test accuracy, yet when used for multi-temporal change detection it fails to agree with the Hansen Global Forest Change reference across three structurally different regions ($F_1 = 0.011$ – 0.251). A diagnostic analysis reveals the cause: on tiles that Hansen records as lost, the classifier labels only 16% as “Forest” in the first year, so a forest-to-non-forest transition is structurally impossible to detect. This stems from domain shift, error multiplication across two noisy classifications, and the absence of a cleared-land class in the source label ontology. Replacing class-transition detection with NDVI differencing—a direct spectral measure of vegetation loss—raises agreement with Global Forest Watch to precision 0.811, recall 0.666, and $F_1 = 0.731$ on a Washington Cascades scene. The study documents where classifier-based change detection breaks down and demonstrates a working alternative.

Index Terms—Deforestation, remote sensing, transfer learning, ResNet-50, EuroSAT, NDVI, change detection, Sentinel-2, Global Forest Watch.

I. INTRODUCTION

Forests regulate the global carbon cycle and harbour a large share of terrestrial biodiversity, and their loss is among the most consequential forms of land-cover change. Timely, spatially explicit detection of deforestation is therefore a priority for environmental monitoring. Freely available satellite archives such as Sentinel-2, together with modern deep learning, make automated detection increasingly feasible.

A widely assumed strategy is to treat deforestation as a by-product of land-cover classification: train a classifier to recognise land-cover types, apply it to imagery of the same region in two years, and mark any tile whose predicted class changes from forest to a human-associated class as deforestation. This is intuitive—if a model can tell forest from

non-forest, running it twice and differencing the results should reveal loss.

This paper tests that assumption and finds it wanting. We train a ResNet-50 classifier on EuroSAT, apply the class-transition strategy across tropical, temperate, and commercial-forestry regions, and validate every detection against the Hansen Global Forest Change dataset that underlies Global Forest Watch. The classifier is highly accurate on its benchmark yet performs poorly at change detection. We diagnose why, and show that NDVI differencing recovers forest loss with substantially higher agreement. Our contributions are: (i) a controlled multi-region demonstration that accurate classification does not imply accurate change detection; (ii) a mechanistic diagnosis of the failure; and (iii) a validated NDVI-based alternative.

II. BACKGROUND

EuroSAT [1] is a Sentinel-2-derived benchmark of 27,000 labelled 64×64 patches across ten land-cover classes. Deep residual networks (ResNet) [2] pre-trained on ImageNet [4] are a standard transfer-learning backbone. For reference-grade monitoring, the Hansen Global Forest Change product [3] provides annual global maps of forest loss at ~ 30 m and underlies Global Forest Watch. The Normalized Difference Vegetation Index (NDVI) [5] measures canopy greenness directly from near-infrared and red bands and is long established for vegetation-change analysis. Sentinel-2 [6] supplies the imagery and Google Earth Engine [7] provides scalable access and compositing. We contrast a classifier-based change-detection pipeline with an index-based one under a common validation protocol.

III. STUDY AREA AND DATA

The classifier is trained on the RGB version of EuroSAT: 27,000 images, ten classes, 64×64 pixels, using a class-stratified 80/20 split (21,600 training, 5,400 test).

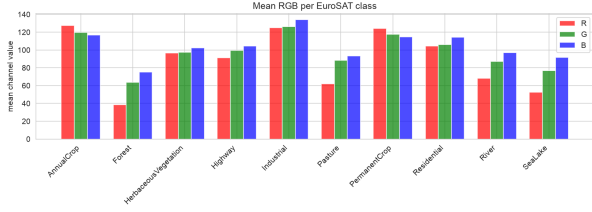


Fig. 1. Mean RGB intensity per EuroSAT class.

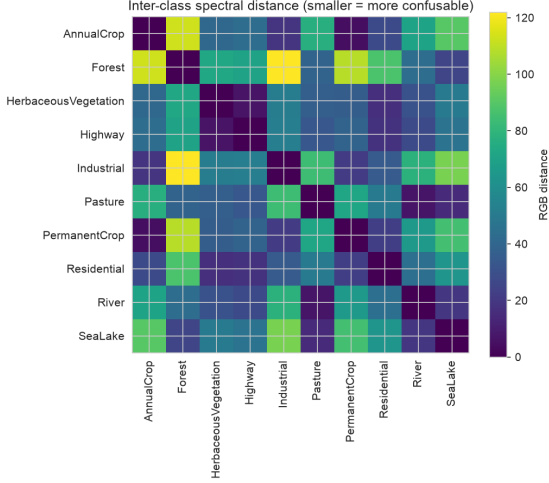


Fig. 2. Inter-class spectral distance (smaller = more confusable).

For change detection we analyse three regions of contrasting forest type and disturbance: the Brazilian Amazon (Rondônia; tropical), the Romanian Carpathians (temperate), and the Washington Cascades, USA (conifer, commercial clear-cut). Detailed validation is reported for the Washington scene, bounding box $[-123.4, 46.8, -122.9, 47.2]$. For each region and each of two years (2018, 2024) we build a cloud-masked ($< 20\%$) summer median composite (1 June–15 September) from Sentinel-2 Surface Reflectance (Harmonized), selecting the red, green, blue, and near-infrared bands at 10 m via Google Earth Engine. The Washington scene is 5567×4457 pixels. Reference forest loss is from Hansen Global Forest Change v1.11 (lossyear band).

A. Exploratory Data Analysis

We first examine both datasets to motivate later design choices. Fig. 1 shows the mean per-channel intensity of each EuroSAT class: Forest is the darkest class and shares a green-dominant, low-red signature with HerbaceousVegetation and Pasture—precisely the classes with which it is later most often confused.

Fig. 2 gives the pairwise RGB distance between class means. Forest is spectrally far from built-up and crop classes but close to the vegetated classes, foreshadowing the misclassification that undermines class-transition change detection.

For the study scene, the NDVI distributions of 2018 and 2024 (Fig. 3) are dominated by a dense-forest peak near 0.9,

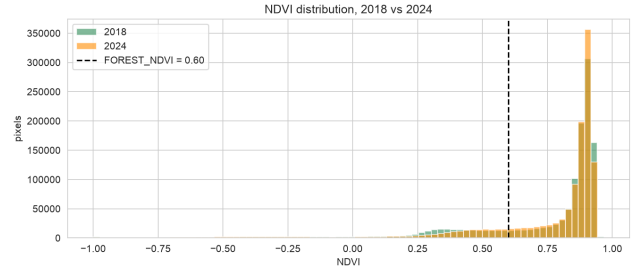


Fig. 3. NDVI distribution, 2018 vs. 2024.

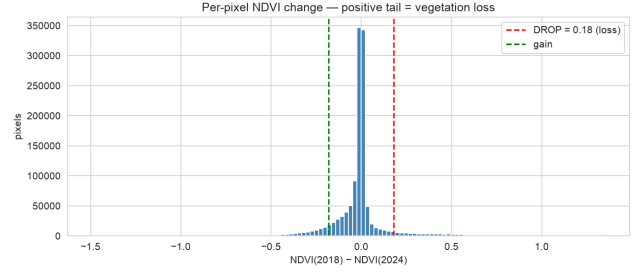


Fig. 4. Per-pixel NDVI change; positive tail = vegetation loss.

which the $\text{NDVI} \geq 0.60$ threshold cleanly isolates. The two distributions are nearly identical, confirming a largely stable-forest scene in which loss is a small signal against a large background.

Per-pixel NDVI change (Fig. 4) is sharply peaked at zero with thin, near-symmetric tails; the $\text{DROP} = 0.18$ threshold lies beyond the noise bulk, justifying its choice and explaining the method’s high precision.

Finally, the annual Hansen loss profile (Fig. 5) shows that the 2018–2024 window captures several of the highest-loss years, yet roughly three-quarters of the region’s total loss pre-dates 2018. Much apparent forest in 2018 is therefore regrowth on older cuts—a source of the false negatives observed later.

IV. METHODOLOGY

A. Land-Cover Classifier

We adopt an ImageNet-pre-trained ResNet-50 [2] and replace its final layer with a two-layer head (a 2048-unit fully connected layer, ReLU, dropout 0.2, ten-way output). Training uses a two-stage schedule: the backbone is first frozen while the head is trained (learning rate 10^{-3}), then the whole network is fine-tuned end-to-end (learning rate 10^{-4}) with a plateau-based schedule and early stopping. We use Adam, cross-entropy loss, ImageNet normalisation, and standard augmentation (flips, rotation, affine shift/shear/zoom).

B. Approach 1: Class-Transition Change Detection

Each composite is partitioned into non-overlapping 64×64 -pixel tiles (640 m). Every tile is classified in both years. A tile is flagged as deforestation when its class is Forest in the earlier year and a non-forest, human-associated class (AnnualCrop,

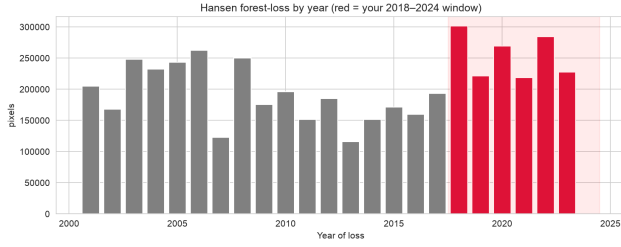


Fig. 5. Hansen forest-loss by year; shaded = 2018–2024 window.

TABLE I
EUROSAT RESNET-50 PER-CLASS TEST PERFORMANCE.

Class	Prec.	Rec.	F_1	Support
AnnualCrop	0.980	0.983	0.982	600
Forest	0.997	0.997	0.997	600
HerbaceousVeg.	0.978	0.983	0.981	600
Highway	0.976	0.988	0.982	500
Industrial	0.994	0.984	0.989	500
Pasture	0.992	0.958	0.975	400
PermanentCrop	0.974	0.976	0.975	500
Residential	0.988	0.998	0.993	600
River	0.980	0.988	0.984	500
SeaLake	1.000	0.995	0.998	600
Accuracy			0.986	5400

Pasture, PermanentCrop, HerbaceousVegetation, Residential, Industrial, or Highway) in the later year.

C. Approach 2: NDVI Differencing

NDVI is computed per 10 m pixel as $(\text{NIR} - \text{Red}) / (\text{NIR} + \text{Red})$. A pixel is loss when it was vegetated in the earlier year ($\text{NDVI} \geq 0.60$) and its NDVI fell by at least DROP. Pixel loss is aggregated to the 640 m grid; a tile is flagged when at least a fraction FRAC of its pixels qualify. DROP and FRAC are the two decision parameters.

D. Validation Protocol

The Hansen `lossyear` raster is reduced to the same 640 m grid: a tile is reference-loss when $\geq 5\%$ of its pixels record loss within 2018–2024. All scene rasters share an identical grid (5567×4457 , EPSG:4326, matching bounds), so detected and reference tiles are directly comparable. Treating Hansen as ground truth, we report true/false positives and negatives, precision, recall, F_1 , and Intersection-over-Union (IoU).

V. RESULTS

A. Land-Cover Classification

The classifier reaches 98.61% overall test accuracy (macro- F_1 0.9855, weighted- F_1 0.9861). The confusion matrix (Fig. 7) is strongly diagonal, and the Forest class is recognised almost perfectly ($F_1 = 0.997$) on the in-domain EuroSAT test set (Table I).

TABLE II
CLASS-TRANSITION CHANGE DETECTION VS. HANSEN, THREE REGIONS.

Region	Forest	Prec.	Rec.	F_1	IoU
Amazon	Tropical	0.194	0.356	0.251	0.144
Carpathians	Temperate	0.318	0.065	0.108	0.057
Cascades	Clear-cut	0.103	0.006	0.011	0.006

TABLE III
CLASSIFIER LABELS ON THE 1,356 HANSEN-LOSS TILES (WASHINGTON).

Predicted class	2018	2024
HerbaceousVegetation	1037	801
Forest	216	511
AnnualCrop	39	12
Industrial	37	5
PermanentCrop	13	15
Highway	12	4
Pasture	1	6
Residential	1	1
SeaLake	0	1

B. Class-Transition Change Detection

Despite the strong classifier, the class-transition strategy agrees poorly with Global Forest Watch in every region (Table II). Performance is uniformly low across tropical, temperate, and clear-cut landscapes—a methodological rather than site-specific failure—and is weakest on the Washington scene, where recall collapses to 0.006.

C. Diagnostic: Why Class-Transition Fails

Hansen records 1,356 tiles as lost in Washington. Table III shows the classifier’s labels on exactly those tiles. Only 216 (16%) were labelled Forest in 2018; most were HerbaceousVegetation. Because a Forest-to-non-forest transition can fire only where the earlier label is Forest, the detector cannot register loss on 84% of truly lost tiles. Worse, the Forest count rises from 216 (2018) to 511 (2024) on these tiles—the classifier drifts the wrong way.

Three compounding factors explain this. First, *domain shift*: EuroSAT’s Forest class represents European forest, and the model frequently misclassifies Pacific-Northwest conifer as herbaceous vegetation, so it cannot recognise the standing forest whose removal it must detect. Second, *error multiplication*: correct change detection requires a correct label in both years, so per-image errors compound and every single-year disagreement registers as spurious change. Third, *ontology mismatch*: EuroSAT has no bare-ground or cleared-land class, so a fresh clear-cut has no correct label to occupy.

D. NDVI Differencing

Replacing class transitions with NDVI differencing removes all three obstacles: NDVI is label-free, a continuous physical measurement rather than a ten-way decision, and captures greenness loss directly. Table IV sweeps the two thresholds against Hansen. Precision stays high (0.81–0.85); recall is the limiting quantity, maximised at the loosest setting. The best configuration (DROP = 0.18, FRAC = 0.05) attains

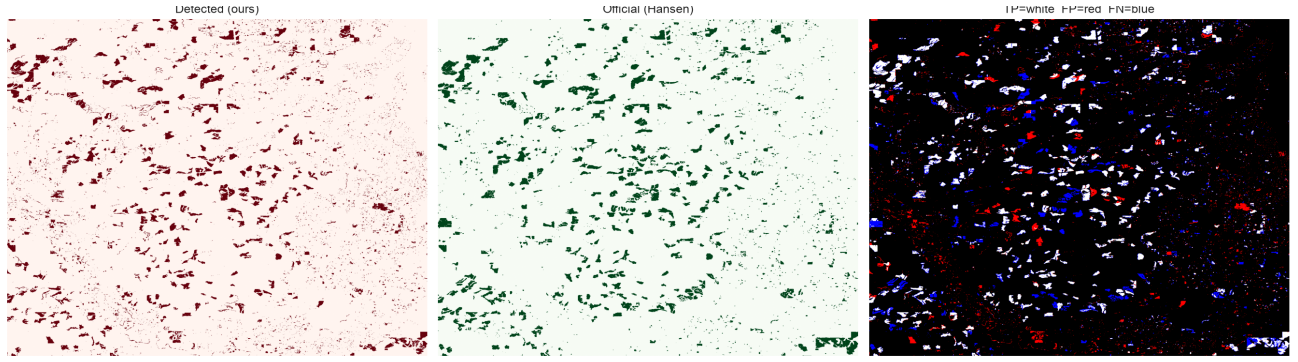


Fig. 6. Validation against Global Forest Watch (Washington). Left: detected loss (ours). Centre: Hansen reference loss. Right: agreement (white = true positive, red = false positive, blue = false negative).

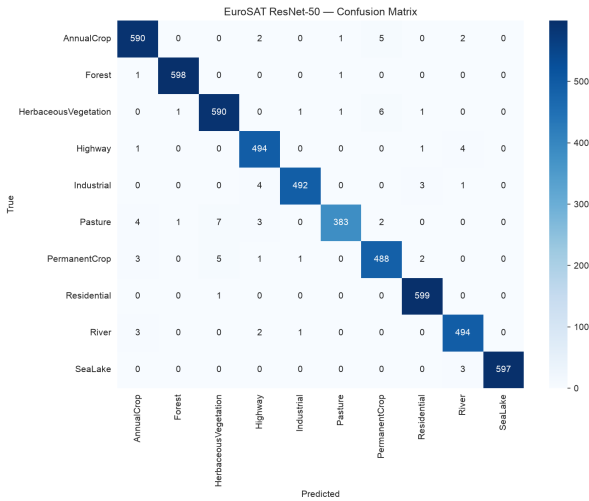


Fig. 7. EuroSAT ResNet-50 confusion matrix on the 5,400-image test set.

TABLE IV
NDVI DIFFERENCING VS. HANSEN (WASHINGTON). BEST ROW IN BOLD.

DROP	FRAC	Prec.	Rec.	F_1	IoU
0.18	0.05	0.811	0.666	0.731	0.576
0.25	0.05	0.838	0.575	0.682	0.518
0.20	0.10	0.854	0.488	0.621	0.451

$F_1 = 0.731$ and $\text{IoU} = 0.576$. The detected-loss map (Fig. 8) shows spatially coherent clusters rather than random speckle.

At the selected operating point the system identifies 903 loss tiles with 211 false positives and 453 false negatives. Fig. 6 compares detected loss, Hansen reference, and their agreement. Relative to class transitions on the same scene, F_1 improves from 0.011 to 0.731—about sixty-fold—confirming that the change signal, not the region, governs performance.

VI. DISCUSSION

The results distinguish two tasks that are easily conflated. Land-cover classification answers “what is here?” and, in-domain, the ResNet-50 answers it with 98.6% accuracy.

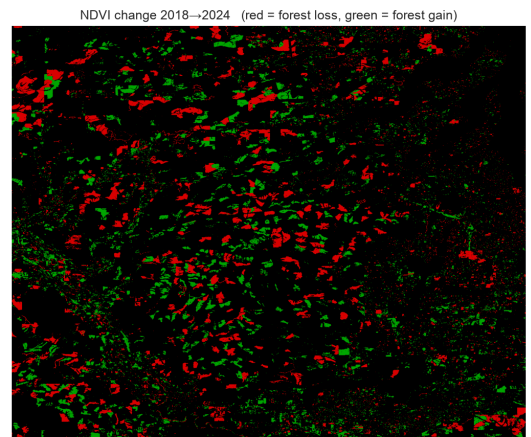


Fig. 8. NDVI change 2018→2024, Washington Cascades (red = forest loss, green = forest gain).

Change detection answers “how much did this location change?”—a different question that a hard, off-domain class label answers poorly. NDVI succeeds because it measures the quantity of interest directly and continuously.

The NDVI sweep shows the expected precision–recall trade-off: stricter thresholds raise precision toward 0.85 but suppress recall. Residual false negatives are dominated by regrowth within the observation window: Pacific-Northwest forestry is fast-cycle, so stands cleared early in 2018–2024 are replanted and recover canopy greenness by 2024, and their endpoint NDVI no longer reflects loss. This is an intrinsic limitation of two-date differencing over a long interval, and explains why recall plateaus near 0.67 while precision stays high.

VII. CONCLUSION

We tested the common assumption that an accurate land-cover classifier can detect deforestation by class differencing across two dates. Across tropical, temperate, and commercial-forestry regions the assumption fails ($F_1 \leq 0.251$), chiefly

because the classifier cannot recognise the target forest off-domain and its label set has no category for cleared land. Substituting NDVI differencing raises agreement with Global Forest Watch to precision 0.811, recall 0.666, and $F_1 = 0.731$ on the Washington Cascades scene. The work contributes both a mechanistically explained negative result and a validated spectral alternative.

VIII. LIMITATIONS AND FUTURE WORK

The 640 m analysis tile is coarse relative to the 30 m Hansen product and cannot resolve narrow logging roads or small clearings; overlapping windows or semantic segmentation would sharpen boundaries. Two-date differencing misses within-window loss that regrows, which a Sentinel-2 time series would recover. NDVI validation is reported for one scene and should extend to the Amazon and Carpathian sites. Adding short-wave-infrared bands (e.g., the Normalized Burn Ratio) would improve discrimination of burned and cleared land.

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